

PALKEM, Sweden

WMO: 021640

Lat: 66.383N

Lon: 21.633E

Elev: 263

StdP: 98.21

Time Zone: 1.00 (EUC)

Period: 83-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.4	-23.4	-28.6	0.3	-26.3	-25.6	0.4	-23.3	12.0	-0.4	10.3	-1.9	0.9	320	0.736

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	22.2	14.4	20.5	13.7	18.7	12.9	15.6	19.9	14.6	18.9	13.7	17.4	3.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	10.3	16.7	12.9	9.6	15.9	11.9	8.9	14.9	44.6	20.0	41.7	19.1	39.2	17.5	19.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	9.5	7.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB													
			WB	-29.8	17.4	4.8	1.4	-33.3	18.4	-36.1	19.2	-38.8	20.0	-42.3	21.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.6	-10.9	-9.4	-6.1	-0.9	5.9	11.0	13.1	11.1	6.0	0.4	-5.9	-8.4
	DBStd	9.62	7.04	6.93	4.57	3.80	3.98	3.55	2.56	3.21	3.25	4.78	5.08	6.10
	HDD10.0	3689	648	544	499	327	142	28	4	26	125	299	477	571
	HDD18.3	6492	906	778	758	577	386	221	162	223	370	557	727	829
	CDD10.0	240	0	0	0	0	15	59	101	61	5	0	0	0
	CDD18.3	1	0	0	0	0	0	1	0	0	0	0	0	0
	CDH23.3	24	0	0	0	0	2	17	3	2	0	0	0	0
	CDH26.7	1	0	0	0	0	0	1	0	0	0	0	0	0
Wind	WSAvg	2.6	2.4	2.8	2.7	2.5	2.7	3.0	2.6	2.5	2.3	2.9	2.4	2.6
Precipitation	PrecAvg	584	44	33	33	28	36	55	72	65	62	52	55	48
	PrecMax	778	76	67	74	69	70	103	122	144	208	96	108	156
	PrecMin	402	1	2	9	8	7	2	20	14	17	18	16	6
	PrecStd	106	18	18	18	16	16	25	28	35	40	20	26	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	5.2	5.9	11.3	22.7	25.5	23.7	23.8	16.1	10.8	5.2	4.4
		MCWB	2.1	2.6	2.2	5.6	13.4	16.2	14.9	17.0	11.5	6.8	3.1	2.2
	2%	DB	1.9	2.9	3.7	8.2	20.1	22.1	21.8	20.6	13.9	9.2	3.3	2.2
		MCWB	-0.3	0.5	0.8	3.6	12.2	14.4	14.1	15.3	10.2	6.6	1.8	0.5
	5%	DB	0.0	1.0	1.8	6.2	16.2	19.8	20.3	18.3	12.2	8.0	1.6	0.9
		MCWB	-1.8	-0.8	-0.4	2.4	9.9	12.9	13.6	13.5	9.4	5.9	0.6	-0.4
	10%	DB	-1.7	-0.6	0.2	4.9	12.9	17.5	18.7	16.8	10.7	6.8	0.6	-0.1
		MCWB	-2.7	-1.6	-1.3	1.6	7.8	11.4	12.9	12.8	8.5	5.2	-0.2	-1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.1	2.8	2.7	6.3	14.1	16.6	16.0	17.4	12.5	9.1	3.5	2.3
		MCDB	4.6	5.2	5.3	9.6	21.0	22.9	20.2	22.0	14.7	9.6	4.7	4.0
	2%	WB	-0.2	0.8	1.0	4.1	12.3	15.1	15.0	16.0	11.3	7.4	2.1	0.8
		MCDB	1.5	2.4	3.1	7.4	19.5	21.4	19.6	18.9	13.1	8.2	3.0	1.7
	5%	WB	-1.5	-0.4	-0.2	2.9	10.2	13.6	14.4	14.7	10.0	6.2	0.8	-0.1
		MCDB	-0.3	0.8	1.4	5.5	15.9	18.8	18.6	17.5	11.2	7.9	1.5	0.7
	10%	WB	-2.8	-1.7	-1.1	1.8	8.2	12.1	13.7	13.3	9.1	5.0	0.0	-0.9
		MCDB	-1.7	-0.5	0.1	4.3	12.5	16.5	17.5	15.9	10.2	6.6	0.6	-0.1
Mean Daily Temperature Range	5% DB	MDBR	5.9	6.3	6.8	7.4	8.4	8.3	7.8	7.2	6.2	5.0	4.9	5.6
		MCDBR	5.7	5.4	7.0	8.7	13.4	12.4	10.8	9.4	7.8	5.9	4.4	6.3
		MCWBR	5.0	4.5	5.4	5.7	7.2	6.4	5.2	5.1	4.8	4.7	4.0	5.9
	5% WB	MCDBR	5.7	5.1	6.0	7.7	12.9	11.1	9.0	7.8	5.8	5.4	3.7	5.4
		MCWBR	5.4	4.3	4.8	5.2	7.0	6.3	4.8	4.8	4.0	4.7	3.5	5.2
Clear-Sky Solar Irradiance	taub	0.185	0.253	0.285	0.316	0.330	0.334	0.345	0.329	0.300	0.277	0.338	N/A	
	taud	1.947	2.157	2.241	2.307	2.480	2.497	2.481	2.510	2.569	2.405	2.742	N/A	
	Ebn at Noon	240	608	778	846	874	877	853	833	783	595	258	N/A	
	Edh at Noon	16	51	79	96	91	91	90	79	59	41	16	N/A	
All-Sky Solar Radiation	RadAvg	0.08	0.64	2.16	4.03	4.95	5.22	4.86	3.59	2.01	0.73	0.15	0.01	
	RadStd	0.01	0.09	0.17	0.20	0.50	0.41	0.47	0.32	0.29	0.12	0.02	0.00	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page